Underreaction, Convexity, and Growth-Optimal Sizing: Harvesting Post-Earnings-Announcement Drift in the Equity and Options Markets

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Abstract

We document, model, and trade post-earnings-announcement drift (PEAD) in two markets at once. On 182,712 analyst-based earnings surprises (1996–2013) the top-minus-bottom surprise-decile spread is $+3.05\%$ over 60 trading days ($t = 5.25$, clustered by announcement quarter). It rises to $+3.76\%$ ($t = 9.25$) after a leave-one-out size and book-to-market adjustment, so the drift is not a size or value premium in disguise. The same ordering shows up in listed options: near-the-money calls on the most positive surprises beat those on the most negative by $+14.38\%$ of premium ($t = 4.69$), and puts show the mirror image (-10.79%, $t = -6.29$). We turn the signal into two books. The first is a sector- and size-neutral, market-hedged equity long/short book. Its holding periods come from an empirical, cell-level decay-time estimator, and it is scaled to its capacity-constrained optimum. It returns 12.15% a year at 12.20% volatility (Sharpe 1.00), with a market beta of -0.015. The second is a long-only options book. For every event it scans the full option chain and picks the contract that maximizes expected logarithmic growth $\mathbb{E}[\log(1 + fR)]$ under a walk-forward, decile-conditioned empirical distribution. It then spreads a fixed premium budget across concurrent bets in proportion to their Kelly fractions. At a 30% outstanding-premium budget it returns 12.26% a year at 3.33% volatility (Sharpe 3.49), and every position's loss is capped at its premium. The two books have a daily return correlation of -0.05. We derive the relevant properties of the selector: its second-order equivalence to Sharpe-ratio ranking, the minimum-stake hurdle that removes lottery-ticket strikes, and the fact that proportional budget allocation is fractional Kelly with an endogenous fraction near 2×10^{-3}. We also derive a capacity bound for the equity book, and we report deflated Sharpe ratios, a full log of the backtest defects we found and fixed, and the limitations that matter for deployment.

Key results (backtest, 1996–2013, $10M initial capital, net of modeled costs)

Book	Ann. ret.	Ann. vol	Sharpe	Max DD	Corr. to mkt
Equity long/short (sector/size-neutral)	12.15%	12.20%	1.00	−36.5%	−0.02
Kelly options (60d, $c = 30\%$)	12.26%	3.33%	3.49	−17.4%	0.20
50/50 blend, 1998–2013[†]	12.73%	6.48%	1.88	−18.5%	–

[†]Illustrative blend computed from the two books' daily NAV series; not a separately backtested strategy. All figures are hypothetical backtest results; see Section 8.

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1 Introduction

Post-earnings-announcement drift is one of the oldest and most persistent anomalies in empirical finance. After a firm reports earnings that differ from expectations, its price does not adjust fully on the announcement. Over the following weeks it keeps moving in the direction of the surprise (Ball and Brown, 1968; Bernard and Thomas, 1989, 1990). The leading explanations are behavioral underreaction and investor inattention (Hirshleifer et al., 2009; DellaVigna and Pollet, 2009), together with limits to arbitrage that keep informed capital from closing the gap quickly (Shleifer and Vishny, 1997; Mendenhall, 2004).

For a practitioner, the drift raises three distinct engineering problems, and this paper takes them in order.

(i) **Identification.** Is the drift in this data a real mispricing, statistically robust to the heavy cross-sectional dependence of earnings seasons? Or is it compensation for size and value exposure? And does it persist in a second, independently priced market (listed options)?

(ii) **Linear harvesting.** Given a noisy, decaying, size-dependent cross-sectional signal, what portfolio isolates it? What holding period does the data support, and where does capacity bind?

(iii) **Convex harvesting.** Options give leverage and bounded loss, but a naive expected-value contract selector is attracted to low-probability, high-payoff strikes. What selection and sizing rule turns a directional edge in the underlying into compounding capital growth?

Contributions. (1) A quarter-clustered event study, a characteristic-adjusted robustness test, and heterogeneity tests establish PEAD in an OptionMetrics-linked universe of 6,766 firms. The same design, applied to option premia, shows the signal carries into options with an amplification of about 4.7×, in line with option elasticity. (2) An equity construction that is dollar-, sector- and size-neutral, with holding periods set by a nonparametric decay-time estimator for each (decile × size × book-to-market) cell. We characterize its capacity frontier analytically and empirically. (3) A growth-optimal options selector. It scores every strike in the day-0 chain against a walk-forward empirical return law, and we prove the properties that make it robust to convexity traps. (4) A joint view: the two books are almost uncorrelated, and in the one regime where the linear book suffers most (the 2009 junk rally), the convex book gains.

Scope. Across the project, eight equity constructions and several options variants were built. This paper treats only the two final designs as "the strategy": the highest-return equity construction, which runs at about 12%, and the Kelly-optimal options book. Earlier variants appear only where they explain a design choice.

2 Data and sample construction

Sources. I/B/E/S analyst consensus and actuals; CRSP daily returns and the CRSP value-weighted index $r_{m,t}$ (`vwretd`); Compustat annual common equity (`ceq`); OptionMetrics IvyDB daily option quotes (about 7.5×10^8 option-day records). The equity universe is every CRSP common stock that had listed options at some point in the window: 8,115 PERMNOs.

Events. An event i is a firm-quarter earnings announcement with a resolved day-0 session (the first trading day at or after the announcement). The surprise is analyst-based. The consensus is restricted to snapshots dated strictly before the announcement (`statpers` $<$ `anndats`), and both actual and forecast are corrected for splits between consensus and announcement. We write

$$\mathrm{SUE}_i \;=\; \frac{A_i - F_i}{s_i}, \tag{1}$$

where A_i is realized EPS, F_i the latest point-in-time consensus, and $s_i > 0$ the firm-specific scale used by the upstream pipeline. Analyst-sourced and seasonal (time-series) surprises sit on different scales (sample s.d. 9.01 vs. 3.36), so they are not pooled. The final sample is 182,712 events on 6,766 firms across $Q = 72$ announcement quarters (Table 1).

Table 1: Sample construction.

Candidate earnings events, 1996–2013	268,606
with resolved day-0 session	261,537
with analyst-based surprise (headline sample)	182,712
with point-in-time book equity and size (characteristic-adjusted sample)	175,016
Linkable to an OptionMetrics `secid` (1:1 links only)	164,673
matched to a near-ATM call or put with ≥ 95 DTE on day 0	126,008
priced by the walk-forward Kelly selector (after 8-quarter warm-up)	113,927

Returns. For event i with day-0 index t_0, the market-adjusted cumulative return to horizon h is $\mathrm{CAR}_i(h) = \prod_{k=1}^{h}(1 + r_{i,t_0+k}) - \prod_{k=1}^{h}(1 + r_{m,t_0+k})$, with $h \in \{1, 5, 10, 20, 40, 60\}$ trading days. Accrual starts on day $t_0 + 1$, so the announcement-day reaction is not counted as drift. This matters for the leakage discussion in Section 7.

3 Establishing the anomaly

3.1 Point-in-time decile assignment

Let $\mathcal{I}_q$ be the events announced in calendar quarter q. Each event gets its within-quarter percentile rank

$$u_i = \frac{1}{|\mathcal{I}_q|}\,\mathrm{rank}_{j\in\mathcal{I}_q}(\mathrm{SUE}_j) \in (0, 1], \qquad D_i = \min\left(\lceil 10\,u_i \rceil, 10\right), \tag{2}$$

with average ranks for ties (SUE rounded to 10^{-8}). Ranking inside the quarter keeps the breakpoints free of multi-year look-ahead. The residual intra-quarter look-ahead is discussed in Section 8. 11.6% of surprises are exactly zero, so D_4 holds 22,423 events against about 15,000–19,000 in the other deciles. That imbalance is a property of the data, not a defect of the sort.

3.2 Inference under earnings-season clustering

Announcements cluster heavily inside each reporting season, and returns of firms announcing in the same season share common shocks. An i.i.d. standard error over 182,712 events would badly overstate precision. We use the Fama–MacBeth estimator (Fama and MacBeth, 1973), taking the

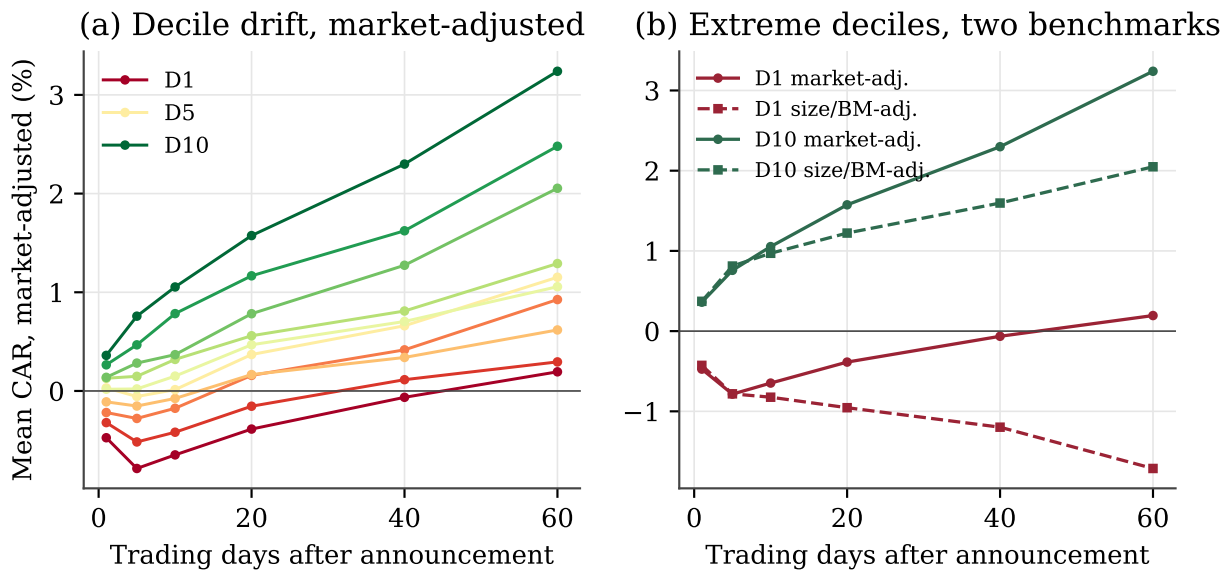


Figure 1: (a) Mean market-adjusted CAR by SUE decile. The ordering is monotone from +10 days onward. (b) Extreme deciles under the market benchmark (solid) and under the leave-one-out size/book-to-market benchmark (dashed). Once size and value are controlled for, D1 declines monotonically to −1.71%.

announcement quarter as the unit of observation. For any event-level quantity x_i and event set $\mathcal{S}$,

$$\bar{x}_q = \frac{1}{|\mathcal{S} \cap \mathcal{I}_q|} \sum_{i \in \mathcal{S} \cap \mathcal{I}_q} x_i, \qquad \hat{\mu} = \frac{1}{Q} \sum_{q=1}^{Q} \bar{x}_q, \qquad \widehat{\mathrm{SE}}(\hat{\mu}) = \frac{1}{\sqrt{Q}} \Big[\tfrac{1}{Q-1} \textstyle\sum_q (\bar{x}_q - \hat{\mu})^2 \Big]^{1/2}. \tag{3}$$

This estimator is consistent under arbitrary within-quarter dependence. The long/short spread is formed *at the quarter level*, $\Delta_q(h) = \overline{\mathrm{CAR}}_{q,D10}(h) - \overline{\mathrm{CAR}}_{q,D1}(h)$, so its standard error reflects the fact that both legs are drawn from the same cross-section.

3.3 Headline result

Table 2 and Figure 1 show the canonical PEAD pattern. The decile means are ordered at every horizon, and from +10 days the ordering is strictly monotone. The spread grows with horizon, from 0.83% at one day to 3.05% at 60 days (Table 3), and every clustered t-statistic exceeds 5.2. A spread that keeps growing is what gradual incorporation of information predicts. A one-off reaction that merely differs by decile would produce a flat spread.

3.4 Risk-based explanations: a leave-one-out characteristic benchmark

In market-adjusted terms, D1 recovers after day 20. One reading is that extreme-bad-news firms are small, high book-to-market firms that earn a size/value premium, which would make part of the "drift" risk compensation. We test this with a characteristic benchmark in the spirit of Daniel et al. (1997). Each event is assigned to a cell $c(i) = (q,\ \text{size quintile},\ \text{BM tercile})$. Book equity comes from Compustat ceq>0 with at least a 130-day reporting lag, both sorts are formed within the

Table 2: Mean market-adjusted CAR (%) by SUE decile and horizon; Fama–MacBeth means over 72 quarters.

Decile	+1d	+5d	+10d	+20d	+40d	+60d
D1 (most negative)	−0.47	−0.79	−0.65	−0.39	−0.06	+0.19
D2	−0.32	−0.52	−0.42	−0.15	+0.11	+0.29
D3	−0.22	−0.28	−0.18	+0.16	+0.42	+0.93
D4	−0.11	−0.15	−0.08	+0.17	+0.34	+0.62
D5	+0.03	−0.06	+0.01	+0.37	+0.66	+1.15
D6	+0.02	+0.02	+0.15	+0.47	+0.70	+1.06
D7	+0.13	+0.15	+0.32	+0.56	+0.81	+1.29
D8	+0.14	+0.28	+0.37	+0.78	+1.27	+2.05
D9	+0.26	+0.47	+0.78	+1.17	+1.62	+2.48
D10 (most positive)	+0.36	+0.76	+1.05	+1.57	+2.30	+3.24

Table 3: D10 − D1 spread by horizon, with quarter-clustered t-statistics. Equity columns use the full event sample; option columns use the option-matched subsample (67 quarters; 65 at +60d) and measure the return on premium of a buy-and-hold near-ATM contract at mid.

	Mkt-adj. equity		Size/BM-adj. equity		Call		Put		Straddle	
Horizon	$\hat{\mu}$	t	$\hat{\mu}$	t	$\hat{\mu}$	t	$\hat{\mu}$	t	$\hat{\mu}$	t
+1d	0.83	10.90	0.80	11.41	4.01	13.33	−2.49	−13.21	0.27	3.31
+5d	1.54	12.23	1.59	14.21	8.03	13.27	−5.11	−13.04	0.77	4.61
+10d	1.70	10.17	1.79	12.30	9.54	11.87	−5.86	−13.01	1.24	4.87
+20d	1.96	8.74	2.18	12.09	11.29	8.34	−6.46	−7.91	2.18	4.97
+40d	2.36	5.64	2.79	9.68	11.76	5.67	−8.27	−6.64	2.80	5.10
+60d	3.05	5.25	3.76	9.25	14.38	4.69	−10.79	−6.29	3.98	5.22

quarter, and 175,016 events are covered. The adjusted return is

$$\widetilde{\mathrm{CAR}}_i(h) = \mathrm{CAR}_i(h) - \frac{\sum_{j \in c(i)} \mathrm{CAR}_j(h) - \mathrm{CAR}_i(h)}{n_{c(i)} - 1}. \tag{4}$$

Leaving the event out stops its own return from shrinking its abnormal return toward zero, an effect that matters in thin cells. After adjustment the spread is *larger* at every horizon (3.76% at 60d, $t = 9.25$), and D1 no longer recovers (Figure 1b). What looked like a D1 reversal was a size/value premium sitting on top of the drift. Adjusting for characteristics removes that premium and exposes the drift in full.

3.5 Heterogeneity and pooling

For grouping variable g we compute the 60-day quarterly spread series $\{\Delta_q^{(g)}\}$ separately within each group and test equality of means with a one-way ANOVA.

- **Sector (FF12):** $F = 0.30$, $p = 0.986$. Point estimates run from 0.98% (Chemicals) to 3.84% (Finance), and every confidence interval overlaps every other. One pooled model across sectors is statistically justified.
- **Size:** $F = 6.50$, $p < 10^{-4}$. The spread falls monotonically from 6.71% ($t = 5.43$) in the smallest quintile to 1.60% ($t = 2.41$) in the largest (Figure 2a). This is the limits-to-arbitrage signature:

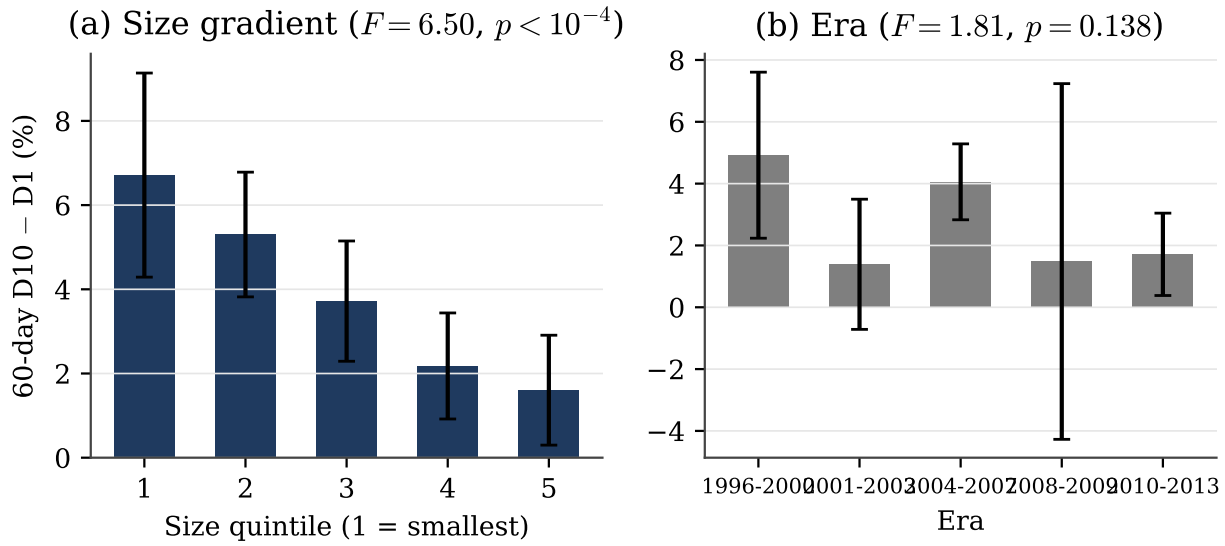


Figure 2: 60-day D10−D1 spread with 95% confidence intervals by (a) size quintile at announcement and (b) era. One-way ANOVA across the groups' quarterly spread series.

the drift is biggest where arbitrage is most expensive. Size therefore must be neutralized in construction and cannot simply be pooled.

- **Era:** $F = 1.81$, $p = 0.138$. Point estimates fall from 4.92% (1996–2000) to 1.4–1.7% after 2008, consistent with post-publication decay (McLean and Pontiff, 2016). The decline is suggestive but not statistically significant.

3.6 Rival hypotheses that were tested and rejected

Surprise reversal. Bernard and Thomas (1990) report a lag-4 SUE autocorrelation of about −0.24. On 141,406 firm pairs we estimate +0.05. Firms in the top decile four quarters earlier still average $\text{SUE}_t = 1.44$, against −0.58 for the bottom decile. The data show persistence, not reversal, so neither book fades a firm's own past surprise.

Announcement-return (EAR) augmentation. Blending the day-0 return into the signal leaves the Sharpe ratio flat between 1.00 and 1.06 across blend weights. That holds once EAR is restricted to information realized *before* the first accrual day. An earlier specification used the day-0-to-day-1 return, which overlaps a position's first day of P&L, and it produced a spurious Sharpe that kept rising with the blend weight. That leak is logged in Section 7.

3.7 Cross-market confirmation: the options market

If PEAD is a real mispricing of the underlying, it should also appear, amplified, in the premium of a leveraged claim on that underlying. For each option-linked event we pick, on day 0, the call (put) with Δ closest to +0.5 (−0.5). The contract needs a two-sided quote and at least 95 calendar days to expiry, and ties are broken toward 120 DTE. We hold it at mid, with no rolling, to each horizon. The inference is the estimator in (3).

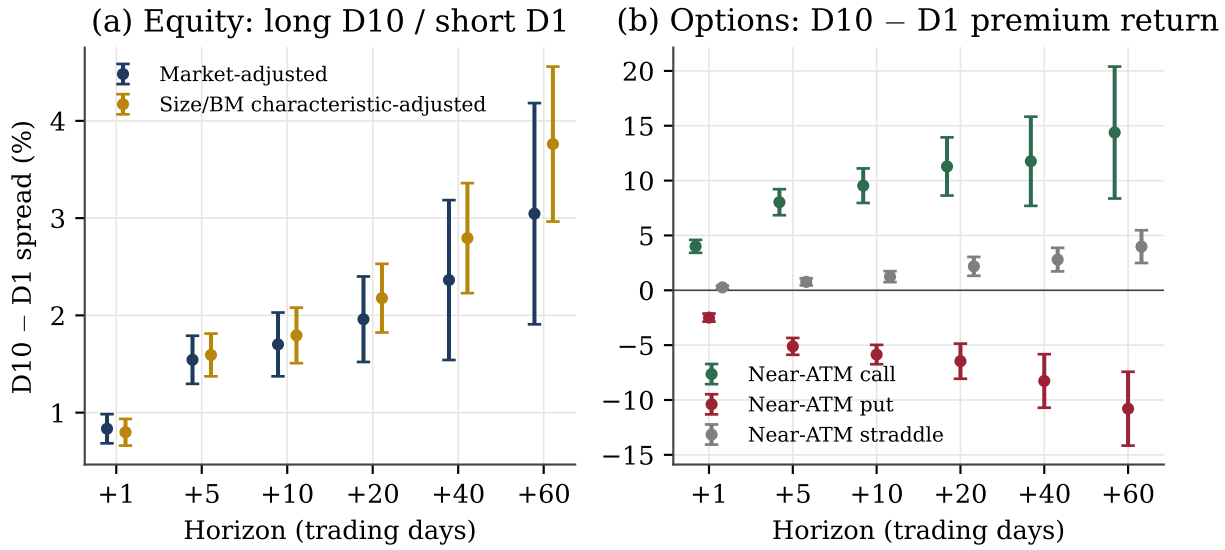


Figure 3: D10−D1 spreads with 95% quarter-clustered confidence intervals. (a) Equity, market-adjusted and characteristic-adjusted. (b) Near-ATM options, as return on premium.

Magnitude. For a call with price C and delta Δ, the Black–Scholes elasticity is $\Omega = \Delta S/C$. At the money $C \approx 0.4\, S\sigma\sqrt{T}$, so $\Omega \approx \Delta/(0.4\,\sigma\sqrt{T})$. With $\Delta \approx 0.5$, $T \approx 1/3$ and $\sigma \in [0.35, 0.45]$, $\Omega \approx 4.8$–6.2. The observed ratio of the 60-day call spread to the stock spread is $14.38/3.05 \approx 4.7$. That sits just below the elasticity band, as expected once theta and vega act on a position held for 60 days. So the options market carries the same directional signal, scaled by leverage (Table 3, Figure 3b).

Direction versus volatility. A long straddle has close to zero delta and so isolates the non-directional component. Its 60-day spread is $+3.98\%$ ($t = 5.22$), about a quarter of the call spread. Extreme-surprise names therefore carry a secondary volatility component. That component is small next to the directional one, which remains the primary effect.

3.8 Decay kinetics and the holding-period estimator

A trading rule has to decide when the marginal day of holding stops paying. Let $\overline{\mathrm{CAR}}_c(d)$ be the Fama–MacBeth mean characteristic-adjusted CAR of holding cell c on day $d = 1, \ldots, 200$. The cells are decile × size × book-to-market. A cell with fewer than 20 quarters or 400 events backs off to decile × size, and then to decile alone; 138 fine and 6 medium cells survive. Daily marginal returns are too noisy to threshold directly, so we use a trailing windowed slope and a cell-specific relative threshold:

$$\varsigma_c(d) = \tfrac{1}{W}\left[\overline{\mathrm{CAR}}_c(d) - \overline{\mathrm{CAR}}_c(d-W)\right], \qquad W = 20, \tag{5}$$

$$\theta_c = 0.15 \max_{d \le 90} |\varsigma_c(d)|, \tag{6}$$

$$\mathcal{E}_c(d) = \left\{|\varsigma_c(d)| < \theta_c\right\} \cup \left\{\operatorname{sgn}\varsigma_c(d) \neq s_c\right\}, \tag{7}$$

$$H_c = \min\left\{d \ge 30 : \textstyle\bigcap_{k=d}^{d+9} \mathcal{E}_c(k)\right\}, \tag{8}$$

where $s_c = +1$ for deciles 6–10 and -1 otherwise. In words, H_c is the first day after a 30-day burn-in on which the drift rate has fallen below 15% of its own peak, or turned the wrong way, and stays there for ten consecutive days. Across cells H_c ranges from 31 to 200 trading days (median 49, mean 57). It is longest in the extreme deciles (mean 63 for D1 and 78 for D10) and shortest near the median (42 for D6).

4 Strategy I: the market-neutral equity book

4.1 Signal-to-weight map

Definition 1 (Trimmed, tilted, doubly-neutral weights)**.** Let $\delta_i = u_i - \frac{1}{2} \in (-\frac{1}{2}, \frac{1}{2}]$, trim $\tau = 0.15$ and tilt $\gamma = 1$. Define the raw conviction

$$t_i = \mathbf{1}\{|\delta_i| > \tau\}\ \operatorname{sgn}(\delta_i)\ \big(2|\delta_i|\big)^{\gamma}.$$

Only the bottom and top 35% of each quarter's surprise distribution are traded. For each quarter q and side $\sigma \in \{+, -\}$, let $\mathcal{G}_{q\sigma}$ be the set of occupied groups $g = (\text{FF12 sector},\ \text{size quintile})$, with $N_{q\sigma} = |\mathcal{G}_{q\sigma}|$. Then

$$\tilde{w}_i = \frac{t_i}{\sum_{j \in g(i)} |t_j|} \cdot \frac{1}{N_{q\sigma}}, \qquad w_i = \frac{\tilde{w}_i}{2\,\overline{|\tilde{w}|}}, \tag{9}$$

where $\overline{|\tilde{w}|}$ is the mean absolute weight over all traded events, a single global constant.

Proposition 1 (Neutrality)**.** *Before capacity caps are applied, the entry cohort of every quarter satisfies: (a) $\sum_{i \in q,\, w_i > 0} w_i = -\sum_{i \in q,\, w_i < 0} w_i$, so the cohort is dollar-neutral; (b) every occupied (sector, size) group receives exactly $1/N_{q\sigma}$ of its side's gross, up to the global constant, and within a group weight is proportional to conviction; (c) the net exposure to sector k is $\big(n_k^+ / N_{q+} - n_k^- / N_{q-}\big)$ times the global constant, where n_k^σ counts the occupied size buckets of sector k on side σ. It is therefore exactly zero whenever both sides occupy the same cells, and the same holds for size.*

The proof is in Appendix A. Neutralization answers a diagnostic finding. Without it, one FF12 sector reached 29% of gross in some quarters, against 8.3% under an even spread. That is unpriced risk unrelated to the PEAD mechanism.

4.2 Sizing, capacity and leverage

Positions enter at the close of day 0 and exit after the holding period $H_{c(i)}$ of their cell. At the start of quarter q, with trailing NAV V_q and base-unit fraction b, the target notional is capped at a fraction κ of 21-day average dollar volume:

$$n_i^{\text{tgt}} = \operatorname{sgn}(w_i)\ \min\big(|w_i|\, b\, V_q,\ \kappa\, \text{ADV}_i\big), \qquad \kappa = 5\%. \tag{10}$$

If the cohort's gross $\sum_{i \in q} |n_i^{\text{tgt}}|$ exceeds $L\, V_q$, positions are funded in descending $|n^{\text{tgt}}|$ until the budget is used up. Weakest convictions are dropped first, instead of every position being shaved proportionally. The headline book uses $b = 0.012$ and $L = 6$.

Table 4: Equity capacity frontier: sweep over the base-unit fraction b at $L = 6$ (all other construction parameters fixed).

b ($\times 10^{-3}$)	Ann. return	Ann. vol	Sharpe	Max DD	% ADV-capped
4.5	8.16%	6.52%	1.24	−22.0%	11.6%
6.0	9.24%	7.90%	1.16	−26.6%	15.5%
9.0	10.89%	10.23%	1.06	−33.3%	22.3%
12.0	**12.15%**	**12.20%**	**1.00**	**−36.5%**	**27.9%**
15.0	10.63%	13.72%	0.81	−43.2%	36.7%
21.0	8.27%	15.25%	0.60	−45.0%	51.8%
36.0	0.39%	22.14%	0.13	−66.2%	62.7%

P&L. Daily P&L is

$$\Pi_t = \sum_{i \text{ open at } t} n_i \left(r_{i,t} - r_{m,t} \right) - \sum_{i \text{ enter/exit at } t} |n_i| \, k_i, \tag{11}$$

where $k_i \in \{25, 15, 10, 7, 5\}$ bp per side by liquidity quintile. Because the return is market-adjusted, each stock position is economically paired with an equal-notional, opposite position in the index. Proposition 1(a) makes these hedges cancel within each cohort, so the residual index hedge is small. The realized market beta is -0.015 (Table 6).

Proposition 2 (Concave capacity frontier)**.** *Assume expected abnormal return per dollar, α_i, is non-increasing in* ADV_i *within each side, as Figure 2a supports. Then the expected alpha per dollar of* marginal *gross, $\partial\, \mathbb{E}[alpha] / \partial\, gross$, is non-increasing in b. Holding volatility fixed, expected arithmetic return is therefore concave in b. After subtracting the variance drag $\frac{1}{2}\sigma^2(b)$, the geometric return has an interior maximum.*

The mechanism is simple. As b grows, the ADV cap binds first on small, illiquid names, which are exactly the names with the largest drift. Marginal capital is pushed into larger, lower-alpha names, while volatility keeps rising roughly in proportion to gross. Table 4 traces the frontier. Annualized return peaks at $b = 0.012$ (12.15%), where 27.9% of positions are ADV-capped. Beyond that point, return falls even as volatility keeps rising. At $b = 0.036$, 62.7% of positions are capped and the return is 0.4%. We chose b by maximizing return subject to Sharpe ≥ 0.6 and maximum drawdown $\geq -40\%$. The drawdown constraint binds first.

Fundamental-law reading. The book holds on average 1,406 positions and opens about 6,600 new ones a year. By the fundamental law of active management (Grinold, 1989), $\text{IR} \approx \text{IC}\sqrt{\text{BR}}$. An information ratio near 1.0 then implies a per-bet information coefficient of only about 0.012. The strategy is a breadth strategy: many weak, nearly independent bets, not a few strong ones.

5 Strategy II: growth-optimal options selection

The near-ATM heuristic of Section 3 keeps one contract per side and discards the rest of the chain. As a traded book it earns 9.51% a year at 9.16% volatility (Sharpe 1.04, maximum drawdown −25.9%). The strategy developed here replaces that heuristic with a selection rule derived from first principles: the growth-optimal (Kelly) criterion (Kelly, 1956; MacLean et al., 2011), applied strike by strike.

5.1 A walk-forward empirical return law

For announcement quarter q and decile d, let $\mathcal{P}_q = \{i : q(i) < q\}$ be the events of all *strictly prior* quarters, with at least 8 quarters of warm-up. Let $\mathcal{P}_{qd} \subset \mathcal{P}_q$ be those in decile d. The market-drift centering term and the conditional shape are estimated separately:

$$\hat{m}_q(h) = \frac{1}{|\mathcal{P}_q|} \sum_{i \in \mathcal{P}_q} \big(R_i^{\text{raw}}(h) - \text{CAR}_i(h)\big), \qquad x_i = \text{CAR}_i(h) + \hat{m}_q(h), \quad i \in \mathcal{P}_{qd}. \tag{12}$$

The pooled sample $\{x_i\}$ is cut at its empirical quantiles into $M = 40$ equal-mass states. State s is represented by its conditional mean, $\rho_{qd,s} = \mathbb{E}[x \mid x \in \text{bucket } s]$, with $p_s = 1/M$. Every cell must contain at least $5M$ observations. Using conditional means instead of bucket edges keeps the tails at their true within-bucket average. Estimating the shape from market-adjusted returns stops the unconditional equity premium of 1996–2013 from swamping the decile-specific tilt. The first build omitted this step and chose calls even in D1.

5.2 Payoff tensor and the growth objective

For event i with spot S_0 on day 0, the candidates are all strikes of the single expiry closest to 120 DTE, subject to DTE ≥ 95, a valid two-sided quote, and at most 60 contracts. Contract k has side $\phi_k \in \{+1, -1\}$ (call or put), strike K_k and mid premium P_k. Its state-contingent return on premium at horizon $h = 60$ is

$$R_{k,s} = \frac{\big(\phi_k \left[S_0(1+\rho_s) - K_k\right]\big)^+}{P_k} - 1 \ \in [-1, \infty). \tag{13}$$

The selector values the option at intrinsic value at the horizon, which is a lower bound on its value because at least 35 days of time value remain. Realized P&L, by contrast, is marked at actual quoted mids. The expected log growth of staking fraction f on contract k is

$$G_k(f) = \sum_{s=1}^{M} p_s \, \log\big(1 + f\, R_{k,s}\big), \qquad g_k^* = \max_{f \in \mathcal{F}} G_k(f), \quad f_k^* = \arg\max_{f \in \mathcal{F}} G_k(f), \tag{14}$$

over the grid $\mathcal{F} = \{0.05, 0.10, \ldots, 0.95\}$. The selected contract is $k^* = \arg\max_k g_k^*$, and the event is traded if and only if $g_{k^*}^* > 0$. Evaluating a batch of events $\times$ strikes $\times$ states $\times$ fractions is one dense tensor reduction, and it runs on a GPU.

Proposition 3 (Well-posedness)**.** *For every $f \in \mathcal{F}$, $1 + fR_{k,s} \geq 1 - f \geq 0.05$, so G_k is finite and no single bet can lose more than the stake. G_k is strictly concave on $[0, 1)$ whenever $R_{k,\cdot}$ is non-degenerate. Moreover $G_k(0) = 0$ and $G_k'(0) = \mathbb{E}[R_k]$, so in the continuum a positive stake is optimal if and only if the contract has positive expected return.*

Proposition 4 (Kelly ranking is Sharpe ranking to second order)**.** *Let $\mu_k = \mathbb{E}[R_k]$, $\sigma_k^2 = \text{Var}(R_k)$ and $m_k = \mathbb{E}[R_k^2] = \sigma_k^2 + \mu_k^2$. A second-order expansion $\log(1+y) = y - \frac{1}{2}y^2 + O(y^3)$ gives*

$$G_k(f) \approx f\mu_k - \tfrac{1}{2} f^2 m_k, \qquad f_k^* \approx \frac{\mu_k}{m_k}, \qquad g_k^* \approx \frac{\mu_k^2}{2m_k} = \frac{1}{2} \cdot \frac{\text{SR}_k^2}{1 + \text{SR}_k^2}, \quad \text{SR}_k = \mu_k/\sigma_k. \tag{15}$$

Among positive-edge contracts, ranking by g^* is therefore asymptotically the same as ranking by the single-bet Sharpe ratio. Naive selection instead ranks by μ_k, which is unbounded in the right tail and ignores σ_k entirely.

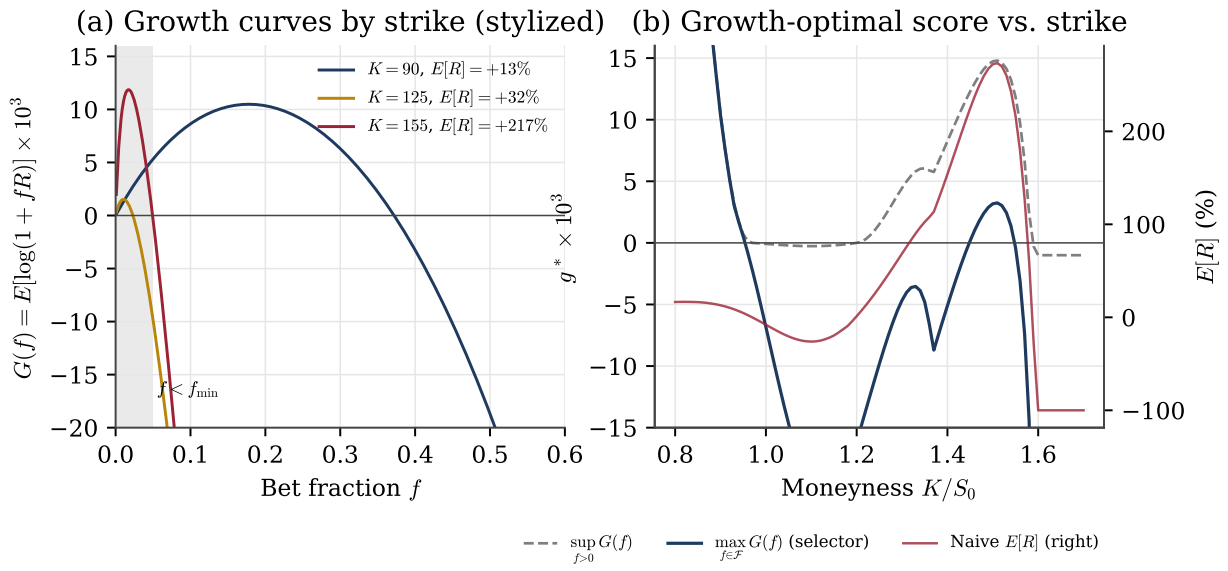


Figure 4: *Stylized illustration, not estimated from data*: a 40-state law with a diffuse body and a 4% right-tail jump, calls priced at 30% implied volatility, 120 DTE, valued at intrinsic at the horizon. (a) Growth curves $G(f)$. The far-OTM strike ($K = 155$) has the largest $E[R]$ and a growth peak at $f \approx 0.02$, inside the excluded region $f < f_{\min}$. (b) Continuous-stake supremum growth (dashed), the selector's grid score (solid) and naive $E[R]$ across strikes.

Proposition 5 (Minimum-stake hurdle)**.** *Because* $\min \mathcal{F} = f_{\min} = 0.05 > 0$*, a contract can be selected only if* $G_k(f_{\min}) > 0$*. To second order this condition reads*

$$\mathbb{E}[R_k] \;>\; \tfrac{f_{\min}}{2}\, \mathbb{E}[R_k^2]. \tag{16}$$

Condition (16) is what actually removes the lottery tickets. Take a far-OTM strike that pays 80× premium with probability 4% and otherwise expires worthless. Then $\mathbb{E}[R^2] \approx 0.04 \cdot 80^2 = 256$, and the hurdle requires $\mathbb{E}[R] > 6.4$, an expected return above 640% of premium. Near-the-money contracts with moderate second moments clear the hurdle with far smaller edges. Figure 4 makes the point on a stylized law. With a continuous stake, the far-OTM strike's supremum growth can match the ITM strike's, because its optimal stake is tiny. On the selector's grid its score collapses. This is a regularizer the method needs, since tail estimates are the least reliable part of any empirical distribution. Quantile bucketing adds a second, milder regularizer: it averages the extreme tail within its bucket.

Validation: the selector finds the trade's direction on its own. Nothing in the selector specifies whether to buy calls or puts. Figure 5 shows that it still recovers the PEAD direction: puts are chosen for 32% of D1 events and 5% of D10 events, and the put share falls almost monotonically in between. Calls dominate unconditionally because the law is centered on the positive trailing market drift. Across picks the mean full-Kelly fraction is 13.8%, and the Kelly pick differs from the naive-EV pick on 29.8% of events.

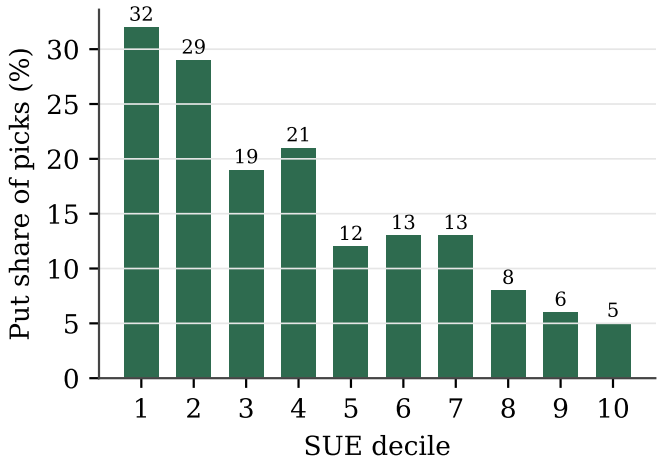


Figure 5: Share of Kelly-optimal picks that are puts, by SUE decile; 113,927 walk-forward-scored events.

5.3 From single-bet Kelly to a portfolio: the premium budget

A typical quarter has about 1,200–1,300 positive-edge candidates. Their single-bet fractions add up to around 170 NAVs, so staking each at f^* is infeasible. Let O_q be the premium still outstanding from earlier cohorts. The quarter's new budget and allocation are

$$B_q = \big(c\,V_q - O_q\big)^+, \qquad \pi_i = B_q\,\frac{h_i}{\sum_{j\in\mathcal{C}_q} h_j}, \qquad h_i = \tfrac{1}{2} f^*_{k^*(i)}, \tag{17}$$

and each position buys $\lfloor \pi_i/(100\,P_i)\rfloor$ contracts. A 3% round-trip cost on premium is charged at entry. Positions are marked at quoted mids on days $\{1,5,10,20,40,60\}$ and exit at day 60. The outstanding-premium test covers *overlapping* cohorts. An earlier implementation applied the budget to each entry cohort alone, which let open premium exceed c and produced an overstated 15.31% return. That result is kept only as a flagged historical comparison (Section 7).

Proposition 6 (Proportional allocation is fractional Kelly)**.** *(a) The half-Kelly factor in* (17) *cancels, so* $\pi_i = \lambda_q f_i^*$ *with* $\lambda_q = B_q/\sum_j f_j^*$*. The book is Kelly-proportional in* relative *sizing, and its absolute scale is set entirely by the budget c. (b) Assume independent bets and the quadratic approximation of Proposition 4. Scaling the vector of single-bet Kelly stakes by* $\lambda \in (0,1]$ *gives portfolio growth* $(2\lambda - \lambda^2)\sum_i g_i^*$ *and a portfolio Sharpe ratio*

$$\mathrm{SR}_{port} = \frac{\sum_i \mu_i^2/m_i}{\big(\sum_i \mu_i^2\sigma_i^2/m_i^2\big)^{1/2}}, \tag{18}$$

which does not depend on λ *and grows as* $\sqrt{N}$ *in the number of bets. (c) Every position's loss is bounded by its premium. The loss on the positions open at any moment is therefore at most c times the NAV at which they were sized.*

With $\sum_j f_j^* \approx 175$ and $c = 0.3$, the book runs at $\lambda \approx 1.7\times10^{-3}$ of full Kelly. It is a deeply fractional Kelly portfolio. Its high Sharpe ratio comes from breadth, as part (b) predicts, not from leverage. The empirical implied information coefficient is $3.49/\sqrt{4{,}770} \approx 0.05$. Part (b) also predicts that return and volatility scale roughly linearly in c while the Sharpe ratio stays approximately

Table 5: Kelly options book: annualized return (Sharpe) by holding horizon and outstanding-premium budget c, with the near-ATM heuristic for comparison.

	+1d	+5d	+10d	+20d	+40d	+60d
$c = 20\%$	−0.01% (−0.02)	1.96% (1.77)	3.49% (1.95)	5.74% (2.49)	8.08% (2.96)	8.58% (3.78)
$c = 30\%$	−0.18% (−0.26)	2.76% (1.62)	5.03% (1.82)	8.41% (2.36)	11.81% (2.84)	**12.26% (3.49)**
$c = 40\%$	−0.36% (−0.39)	3.57% (1.55)	6.59% (1.76)	11.08% (2.30)	15.62% (2.79)	15.79% (3.28)
Near-ATM	−1.01% (−0.78)	0.78% (0.31)	2.13% (0.60)	2.91% (0.60)	4.67% (0.72)	9.51% (1.04)
Max DD, +60d	$c = 20\%$: −10.3%; $c = 30\%$: −17.4%; $c = 40\%$: −23.9%; near-ATM: −25.9%					

Table 6: Risk profile of the two books. Summary statistics come from the committed backtest summaries. Rows marked ‡ are computed from the daily NAV series over each book's *active* period (equity from 1996-01; options from 1998-01).

	Equity book	Options book ($c = 30\%$, 60d)
Annualized return (full window)	12.15%	12.26%
Annualized volatility	12.20%	3.33%
Sharpe ratio	1.00	3.49
Maximum drawdown (trough date)	−36.5% (2010-05-07)	−17.4% (2009-03-11)
Final NAV from \$10M	\$93.3M	\$68.8M
CAGR, active period‡	12.85%	14.06%
Sharpe, active period‡	1.03	3.73
Sortino‡	1.39	5.88
Skewness / kurtosis of daily returns‡	−0.22 / 10.1	+0.99 / 11.0
Daily 99% VaR / CVaR‡	−2.13% / −3.05%	−0.48% / −0.70%
Market beta / correlation‡	−0.015 / −0.02	0.034 / 0.20
CAPM alpha, annualized (t-stat)‡	13.0% (4.47)	12.9% (14.26)
Calmar ratio‡	0.35	0.81
Positions	119,130 (avg. 1,406 open)	74,926 sized of 81,605 eligible
Realized gross / NAV‡	avg. 4.5×, max 7.6×	max open premium 29.3% of NAV
Share of positions at ADV cap	27.9%	–

unchanged. Table 5 confirms this: going from $c = 20\%$ to 30% to 40% moves return from 8.58% to 12.26% to 15.79% and volatility from 2.18% to 3.33% to 4.50%, while Sharpe decays only mildly (3.78, 3.49, 3.28). Compounding, whole-contract rounding and the overlap constraint account for the decay.

Returns increase with horizon, which is exactly the decay profile of Section 3.8: a 3% round-trip cost cannot be earned back from a one-day drift of 0.8%, but it can be from 60 days of accumulated drift. Replacing the heuristic with the Kelly selector at equal horizon triples the Sharpe ratio and cuts the drawdown by more than half, largely because the heuristic funded only 2,007 of 99,294 signals, while the proportional allocation spreads the budget over all of them.

6 Results

6.1 Standalone performance

Table 6 and Figure 6 summarize the two books. They reach similar returns through very different risk profiles. The equity book is a linear, market-hedged long/short: negative skew, fat tails, a

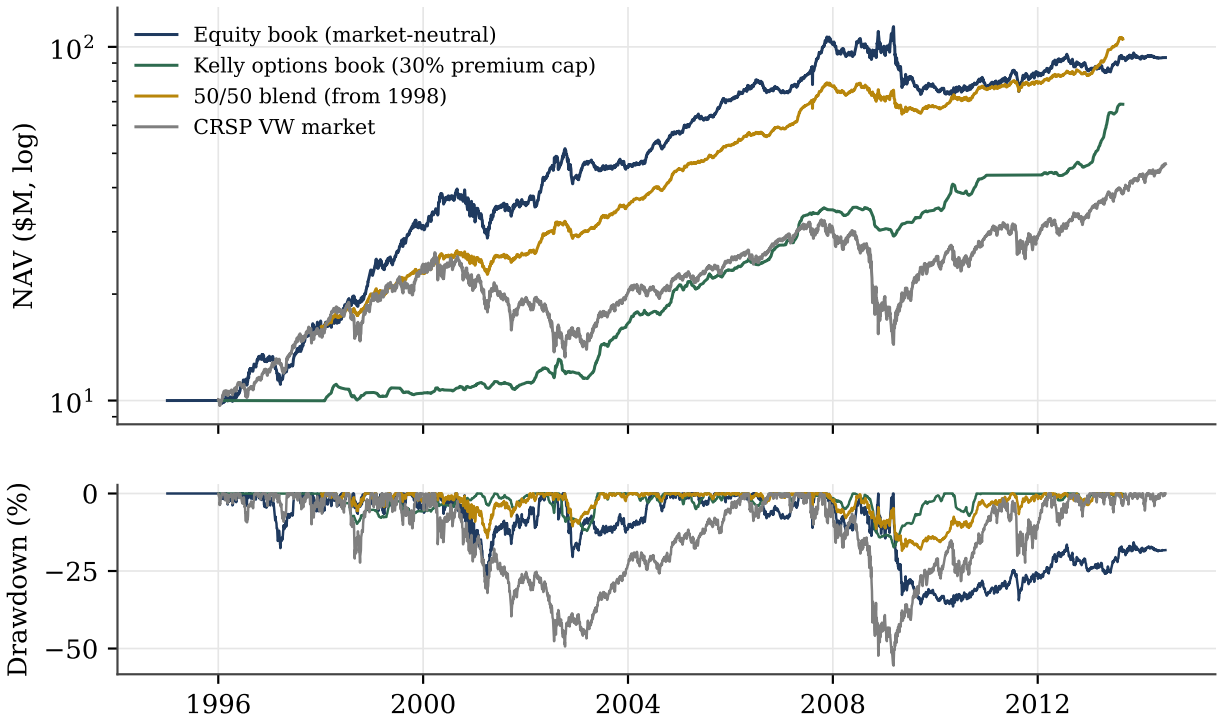


Figure 6: NAV (log scale, $10M start) and drawdowns. The options book is flat through its 8-quarter walk-forward warm-up (1996–1997). The blend begins in 1998.

−36.5% drawdown, and no market beta. The options book is convex and budget-constrained: positive skew, daily tail losses about four times smaller, and a structural loss bound (Proposition 6c). It carries a modest market correlation of 0.20, which comes from its unconditional call bias. For both books the CAPM alpha is essentially the whole return.

6.2 Complementarity

The two books are almost orthogonal: daily correlation −0.055 and monthly correlation −0.12 over 175 months. Calendar-year returns (Table 7, Figure 7) show why. In 2009 the equity book lost 23.2%, its worst year, while the options book gained 14.3%. The 2009 rebound was a rally in low-quality, heavily shorted names, the same regime that produces momentum crashes (Daniel and Moskowitz, 2016). A linear short book pays for such a rally without limit. A long put pays at most its premium, and long calls on positive-surprise names take part in the rally.

Table 8 reports static blends, rebalanced daily with zero rebalancing cost, over the options book's active window. At 50/50 the blend keeps the return of either book (12.7%), roughly doubles the equity book's Sharpe ratio (0.94 to 1.88 in this window) and halves its drawdown. The in-sample tangency portfolio puts about 92% on the options book. We do not recommend that allocation: the options book's capacity and cost assumptions (Section 8) are much less certain than its Sharpe ratio suggests, and the equity book diversifies model risk as well as market risk.

Table 7: Calendar-year returns (%). The options book has no data for 2011: that year's OptionMetrics price file is corrupt in the source extract. Its 2012 figure includes positions whose holding windows straddled the gap, and its 2013 figure covers January to August only.

Year	Equity	Options	Market	Year	Equity	Options	Market
1996	33.1	– (warm-up)	21.3	2005	23.0	9.7	7.3
1997	26.1	– (warm-up)	30.4	2006	5.0	17.8	16.2
1998	37.1	5.4	22.3	2007	37.3	25.9	7.4
1999	35.9	−0.5	25.7	2008	−1.6	−12.0	−38.2
2000	12.6	4.9	−11.3	2009	−23.2	14.3	31.2
2001	2.8	2.2	−11.0	2010	−0.1	24.5	17.7
2002	18.7	6.4	−20.9	2011	9.6	n/a	−1.1
2003	6.1	38.5	33.2	2012	2.2	7.1	15.8
2004	25.9	28.5	13.0	2013	7.4	48.3 (Jan–Aug)	30.5

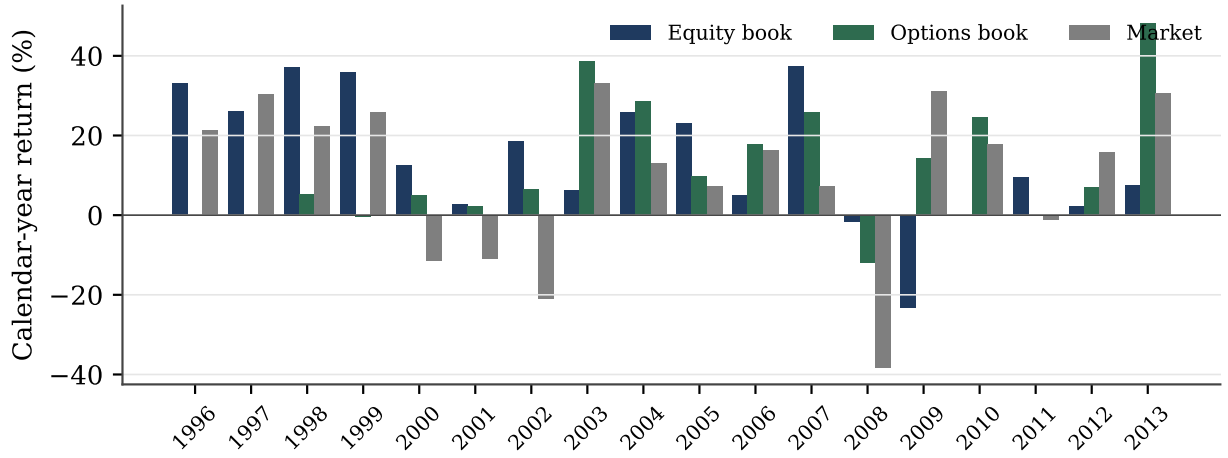


Figure 7: Calendar-year returns of both books against the CRSP value-weighted market.

6.3 Capacity and risk-budget frontiers

Figure 8 shows the main difference between the two books as scalable products. Equity return is capacity-bound: the frontier bends over at a finite b (Proposition 2). Options return is budget-bound: to first order it scales linearly with c, and c is a risk-appetite dial with a hard loss interpretation. The practical limits on the options book lie outside the backtest: quoted width, depth and market impact in single-name options.

7 Statistical robustness and backtest hygiene

7.1 Multiple testing

Both books were selected from a family of trials, so we deflate their Sharpe ratios (Bailey and López de Prado, 2014). For N trials whose annualized Sharpe ratios have variance V, the expected maximum under the null is

$$\widehat{\mathrm{SR}}_0 = \sqrt{V}\left[(1-\gamma_E)\,\Phi^{-1}\!\left(1-\tfrac{1}{N}\right) + \gamma_E\,\Phi^{-1}\!\left(1-\tfrac{1}{Ne}\right)\right], \tag{19}$$

Table 8: Static blends, 1998-01-06 to 2013-08-30 (3,939 days; options returns set to zero on dates absent from its calendar).

Equity weight	CAGR	Vol	Sharpe	Max DD
100%	11.82%	12.68%	0.94	−36.5%
75%	12.33%	9.50%	1.27	−25.6%
50%	12.73%	6.48%	1.88	−18.5%
25%	12.99%	3.98%	3.09	−15.0%
0%	13.13%	3.44%	3.61	−17.4%

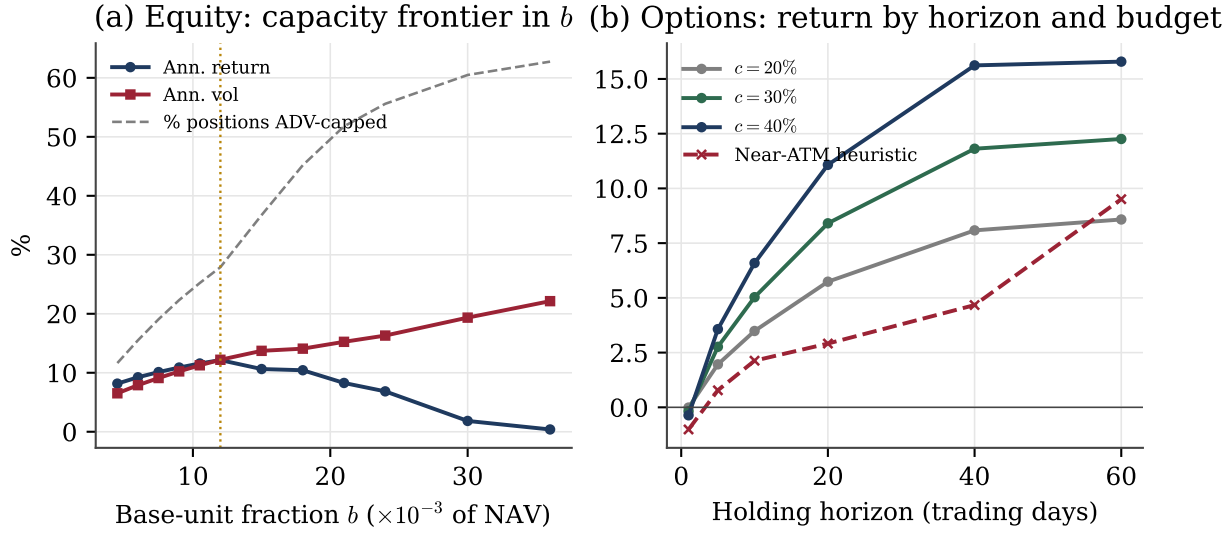


Figure 8: (a) Equity: return, volatility and the share of ADV-capped positions against the base-unit fraction b. The dotted line marks the chosen $b = 0.012$. (b) Options: annualized return against holding horizon for three premium budgets and for the near-ATM heuristic.

where γ_E is the Euler–Mascheroni constant. The deflated Sharpe ratio is the probabilistic Sharpe ratio (Bailey and López de Prado, 2012) measured against $\widehat{\mathrm{SR}}_0$, allowing for the skewness and kurtosis of the returns:

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \widehat{\mathrm{SR}}_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right), \tag{20}$$

with Sharpe ratios in per-period units. The results are:

- **Equity:** $N = 93$ (every point of the aggressive sizing sweep across all designs), $\widehat{\mathrm{SR}}_0 = 0.66$, $\mathrm{DSR} = 0.94$. The book beats the best-of-93 null with 94% probability, just short of the conventional 95%. A committee should read the equity book's Sharpe ratio as *probably but not conclusively* greater than what selection alone would produce. The signal behind it is unambiguous (t-statistics of 5 to 14 in Section 3); the uncertainty is about how well the portfolio's calibration generalizes.
- **Options:** $N = 24$ (four constructions × six horizons), $\widehat{\mathrm{SR}}_0 = 2.56$, $\mathrm{DSR} > 0.9999$.

These trial counts are lower bounds. Earlier exploratory sweeps (trim, tilt, leverage cap) widen the

Table 9: Backtest defects detected and corrected during development.

Defect	Symptom that exposed it	Effect of the fix
Sharpe denominator	Daily returns divided by initial rather than prior-day NAV, overstating volatility as NAV compounded	Sharpe ratios rose; returns and drawdowns unchanged
EAR leakage	Day-0-to-day-1 return overlapped the first day of held P&L; Sharpe rose monotonically with EAR weight	Sharpe flat at 1.00–1.06; EAR not adopted
Ambient beta in return law	Selector chose calls even in D1 (1996–2013 was a net up market)	Shape from market-adjusted returns, re-centered on trailing drift
OptionMetrics strike units	Strikes stored $\times 1000$; puts dominated even D10 with Kelly fractions near 1 (a "free money" artifact)	Strikes divided by 1000 at source
Full-sample return law	1998 trades priced with 2013 data	Walk-forward law, 8-quarter warm-up; Sharpe about 4.0 $\rightarrow$ 3.56
Independent Kelly sizing	Budget exhausted by a few bets; 46 positions funded in 17 years	Kelly-proportional allocation of a shared budget
Per-cohort premium cap	Open premium could exceed c across overlapping 60-day cohorts	Cap on total outstanding premium; 15.31% $\rightarrow$ 12.26% (at $c = 30\%$)
Data drift on a staged copy	40% of decay-day cells changed with no code change	Results restored to canonical data; checksum manifest and pre-commit hook

family. Even doubling N moves $\widehat{\mathrm{SR}}_0$ only by roughly the change in $\Phi^{-1}(1 - 1/N)$, from about 2.3 to 2.6 standard deviations.

7.2 Audit log: defects found and corrected

Table 9 lists every defect the development process caught. We report them because the *direction* of each correction shows how fragile each headline number is. Every correction lowered or left unchanged the reported performance.

We also inspected the day-level P&L of the options book. Its worst days fall on 2008-10-24, 2008-11-20 and September 2002, which are real stress episodes, not artifacts of a smoothed P&L.

8 Limitations and implementation risk

We list these without softening. Each item is either a reason realized performance would be lower or a test still to be run.

1. **Sample period.** All results are for 1996–2013, and the era analysis shows a weakening drift after 2008. A data extension through the present and a post-publication decay adjustment (McLean and Pontiff, 2016) are required before any capital is allocated.

2. **Intra-quarter ranking.** Deciles are formed from each announcement quarter's full cross-section, so an early-quarter event is ranked partly against announcements that come later. This is a mild look-ahead that the literature commonly accepts. A tradeable version must rank against trailing breakpoints, for example the previous quarter's SUE distribution.

3. **In-sample calibration of the equity book.** The decay-day holding periods, the trim and tilt, and the base-unit fraction were all chosen on the full sample. The DSR of 0.94 reflects this; a walk-forward re-estimation of H_c and b is the obvious next test. (A walk-forward ML extension, "Strategy 8", exists but ran on an unverified data copy and is excluded here.)

4. **Equity costs and leverage.** Costs are 5–25 bp per side by liquidity quintile. Borrow fees, index-hedge costs and financing are not modeled. At the chosen sizing, 27.9% of positions sit at 5% of ADV, where impact is likely to exceed the modeled cost (Frazzini et al., 2018; Novy-Marx and Velikov, 2016). The gross cap $L = 6$ applies per entry cohort. Across overlapping cohorts, realized gross averages 4.5× NAV and peaks at 7.6×, which is prime-brokerage-level leverage.

5. **Option execution.** Entries and marks use mids with a flat 3% round-trip charge. Quoted single-name spreads, especially on OTM strikes and in the early sample, are often wider. Missing forward quotes are forward-filled. The book opens about 4,800 small positions a year; capacity and impact have not been modeled. Replacing the flat charge with the actual quoted bid/ask on the selected contracts is the single most important robustness test outstanding.

6. **Options data coverage.** 2011 is missing because its source file is corrupt, so the trading-day calendar compresses across the gap, and 2013 ends in August. The matched sample is tilted toward names with liquid, longer-dated chains.

7. **Return-law pooling.** One distribution per (decile, quarter) is shared by every stock in the decile, regardless of its own volatility. Scaling the states by each stock's volatility (or implied volatility) should sharpen strike selection and is not yet built.

8. **Universe.** Only stocks with listed options at some point in the window are included. Micro-caps with no options, where PEAD is historically strongest, are excluded. That makes the equity results conservative in magnitude but relevant to a liquid mandate.

9 Conclusion

PEAD in this sample passes the tests an allocator should require. It is significant under earnings-season clustering. It survives, and grows under, a size/value characteristic benchmark. It is homogeneous across industries, graded by size as limits to arbitrage predicts, and confirmed in a second, independently priced market. We harvest it in two ways that are complementary in both mathematics and risk. The linear book neutralizes every unpriced exposure we could measure and scales to its capacity optimum, which yields about 12% a year with no market beta. The convex book treats each event as a bet on an empirical distribution, picks the strike that maximizes expected log growth, and sizes the book as a deeply fractional Kelly portfolio under a hard premium budget, which also yields about 12% a year, with a Sharpe ratio driven by breadth and losses bounded by construction. The two books are close to orthogonal and hedge each other's worst regime, so they are best considered together. The work that remains before deployment is concrete: data after 2013, trailing-breakpoint ranking, walk-forward calibration of the equity book, and quote-level option execution costs.

Important disclosure. All performance figures in this paper are hypothetical and come from historical backtests. They do not reflect actual trading, and they benefit from hindsight in model design. Backtested results have inherent limitations, including the modeling assumptions in Section 8; realized results may differ materially. Nothing here is an offer, a solicitation, or investment advice.

A Proofs

Proof of Proposition 1. Fix q and side σ. For an occupied group g, $\sum_{i\in g}|\tilde{w}_i| = \sum_{i\in g}|t_i| / \big(\sum_{j\in g}|t_j| \cdot N_{q\sigma}\big) = 1/N_{q\sigma}$, which proves (b). Summing over the $N_{q\sigma}$ groups of the side gives $\sum_{i\in q\sigma}|\tilde{w}_i| = 1$ for both sides. Every weight on a side has the same sign, so $\sum_{w>0}\tilde{w} = 1 = -\sum_{w<0}\tilde{w}$. The global rescaling in (9) multiplies every weight by the same positive constant, which proves (a). For (c), sector k's long gross is the sum over its occupied long groups, n_k^+/N_{q+}, and likewise on the short side. Net exposure is the difference, and it vanishes when $n_k^+/N_{q+} = n_k^-/N_{q-}$, in particular when both sides occupy the same set of cells. The size argument is symmetric. □

Proof of Proposition 2. Fix a cohort. By (10), $\partial n_i/\partial b = |w_i|V_q\,\mathbf{1}\{|w_i|bV_q < \kappa \mathrm{ADV}_i\}$. Position i stops absorbing marginal capital at $b_i = \kappa\mathrm{ADV}_i/(|w_i|V_q)$, and this threshold increases in ADV_i. Write $\mathcal{U}(b) = \{i : b < b_i\}$ for the uncapped set, which shrinks as b grows. The marginal alpha per marginal gross dollar is the $|w|$-weighted average of α_i over $\mathcal{U}(b)$. Raising b removes the members with the smallest b_i, which by the monotonicity of b_i in ADV are the smallest names, and hence, under the assumption, those with the largest α_i. The weighted average therefore cannot increase. Expected arithmetic P&L, the integral of this non-increasing marginal alpha, is concave in b. Volatility is non-decreasing in gross, so $\mu(b) - \frac{1}{2}\sigma^2(b)$ is concave with $\mu(0) = \sigma(0) = 0$. Its derivative at zero is positive when the uncapped book has positive alpha. As $b \to \infty$, alpha saturates while the variance term does not, so the function eventually decreases, and the maximum is interior. □

Proof of Proposition 3. $R_{k,s} \geq -1$ and $f \leq 0.95$ give $1 + fR_{k,s} \geq 0.05$. The second derivative is $G_k''(f) = -\sum_s p_s R_{k,s}^2/(1+fR_{k,s})^2 < 0$ unless every $R_{k,s} = 0$. Direct evaluation gives $G_k(0) = 0$ and $G_k'(0) = \sum_s p_s R_{k,s}$. Since G_k is concave, its continuum maximizer on $[0,1)$ is positive if and only if $G_k'(0) > 0$. □

Proof of Proposition 4. Substitute the expansion of the logarithm and take expectations, which gives $G_k(f) = f\mu_k - \frac{1}{2}f^2 m_k + O(f^3\mathbb{E}|R_k|^3)$. Maximizing the quadratic gives $f^* = \mu_k/m_k$ and $g^* = \mu_k^2/(2m_k)$. Dividing numerator and denominator by σ_k^2 gives $\frac{1}{2}\mathrm{SR}_k^2/(1+\mathrm{SR}_k^2)$, which is increasing in SR_k for $\mathrm{SR}_k > 0$. □

Proof of Proposition 5. The selected contract maximizes $g_k^* = \max_{f\in\mathcal{F}} G_k(f)$ and is traded only if that maximum is positive. G_k is concave with $G_k(0) = 0$. If $G_k(f_{\min}) \leq 0$, then concavity gives $G_k(f) \leq (f/f_{\min})\,G_k(f_{\min})$ for every $f \geq f_{\min}$, so $G_k(f) \leq 0$ on all of $\mathcal{F}$. Selectability therefore requires $G_k(f_{\min}) > 0$. In the quadratic approximation this is $f_{\min}\mu_k - \frac{1}{2}f_{\min}^2 m_k > 0$, which is (16). □

Proof of Proposition 6. (a) $h_i / \sum_j h_j = f_i^* / \sum_j f_j^*$. (b) Under independence and the quadratic approximation, portfolio growth is $\sum_i (\pi_i \mu_i - \frac{1}{2}\pi_i^2 m_i)$. Setting $\pi_i = \lambda \mu_i / m_i$ gives

$$\sum_i \frac{\mu_i^2}{m_i}\left(\lambda - \tfrac{1}{2}\lambda^2\right) = (2\lambda - \lambda^2)\sum_i g_i^*.$$

The portfolio mean is $\lambda \sum_i \mu_i^2 / m_i$ and its standard deviation is $\lambda(\sum_i \mu_i^2 \sigma_i^2 / m_i^2)^{1/2}$, so λ cancels in their ratio. With N bets of comparable size the numerator is $O(N)$ and the denominator $O(\sqrt{N})$. (c) A long option position costs its premium and has a non-negative payoff, so its loss is at most the premium. Premium outstanding is at most cV_q by construction of B_q. □

B Parameter summary and reproducibility map

Equity book	
Signal	Within-quarter percentile of analyst SUE; trim $\tau = 0.15$, tilt $\gamma = 1$
Neutrality	Dollar-neutral per cohort; equal gross per (FF12 sector × size quintile) group per side
Holding period	Cell decay day H_c (Section 3.8); 31–200 trading days
Sizing	$b = 0.012$ of trailing NAV per unit weight; ADV cap 5% of 21-day dollar volume; cohort gross cap 6× NAV with priority trimming
Costs	25/15/10/7/5 bp per side by liquidity quintile
Options book	
Universe	All strikes, single expiry closest to 120 DTE (≥ 95), up to 60 contracts per event
Return law	Walk-forward, decile-conditioned, 40 equal-mass quantile states, 60-day horizon, 8-quarter warm-up
Selection	$\arg\max_k \max_{f \in \{0.05,\ldots,0.95\}} \mathbb{E}[\log(1 + fR_k)]$; trade iff $g^* > 0$
Sizing	Outstanding-premium budget $c = 30\%$ of trailing NAV, allocated in proportion to f^*
Holding / costs	60 trading days, marked at quoted mids; 3% round-trip on premium

Code. Evidence: `scripts/equity_pead/01--10`; options evidence: `scripts/options_pead/33--39`; equity book: `scripts/equity_strategy/14--16` (decay and positions), `24` (construction), `36` (sizing frontier); options book: `scripts/options_strategy/46` (return law), `47/47b` (chains), `48` (selector), `49--50` (pricing and backtest). The supplementary statistics in this paper (risk moments, DSR, blends, calendar returns) and all figures come from `scripts/build_quant_paper_figures.py`, which writes `reports/paper/paper_stats.json`.

References

Bailey, D. H., and M. López de Prado (2012). The Sharpe ratio efficient frontier. *Journal of Risk* 15(2), 3–44.

Bailey, D. H., and M. López de Prado (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management* 40(5), 94–107.

Ball, R., and P. Brown (1968). An empirical evaluation of accounting income numbers. *Journal of Accounting Research* 6(2), 159–178.

Bernard, V. L., and J. K. Thomas (1989). Post-earnings-announcement drift: delayed price response or risk premium? *Journal of Accounting Research* 27 (Supplement), 1–36.

Bernard, V. L., and J. K. Thomas (1990). Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13(4), 305–340.

Daniel, K., M. Grinblatt, S. Titman, and R. Wermers (1997). Measuring mutual fund performance with characteristic-based benchmarks. *Journal of Finance* 52(3), 1035–1058.

Daniel, K., and T. J. Moskowitz (2016). Momentum crashes. *Journal of Financial Economics* 122(2), 221–247.

DellaVigna, S., and J. M. Pollet (2009). Investor inattention and Friday earnings announcements. *Journal of Finance* 64(2), 709–749.

Fama, E. F., and J. D. MacBeth (1973). Risk, return, and equilibrium: empirical tests. *Journal of Political Economy* 81(3), 607–636.

Frazzini, A., R. Israel, and T. J. Moskowitz (2018). Trading costs. Working paper, AQR Capital Management.

Grinold, R. C. (1989). The fundamental law of active management. *Journal of Portfolio Management* 15(3), 30–37.

Hirshleifer, D., S. S. Lim, and S. H. Teoh (2009). Driven to distraction: extraneous events and underreaction to earnings news. *Journal of Finance* 64(5), 2289–2325.

Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal* 35(4), 917–926.

MacLean, L. C., E. O. Thorp, and W. T. Ziemba, eds. (2011). *The Kelly Capital Growth Investment Criterion: Theory and Practice*. World Scientific.

McLean, R. D., and J. Pontiff (2016). Does academic research destroy stock return predictability? *Journal of Finance* 71(1), 5–32.

Mendenhall, R. R. (2004). Arbitrage risk and post-earnings-announcement drift. *Journal of Business* 77(4), 875–894.

Novy-Marx, R., and M. Velikov (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies* 29(1), 104–147.

Shleifer, A., and R. W. Vishny (1997). The limits of arbitrage. *Journal of Finance* 52(1), 35–55.